

Sta 610: Case Study

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Abstract

This study is going to explore the relationship between the price per mg of diazepam and other factors related to the sale using the StreetRx data. We are going to answer these research questions:

- Which variables are associated with pricing per milligram of Diazepam?
- Is there heterogeneity in pricing of Diazepam by location?

Data Cleaning

Missing Data Since ppm is our response variable, we remove all observations with missing ppm values, as it is nonsensical to include observations without a response.

The `primary_reason` variable has over 50% missing observations, making it unsuitable for modeling. We therefore exclude this variable from our analysis.

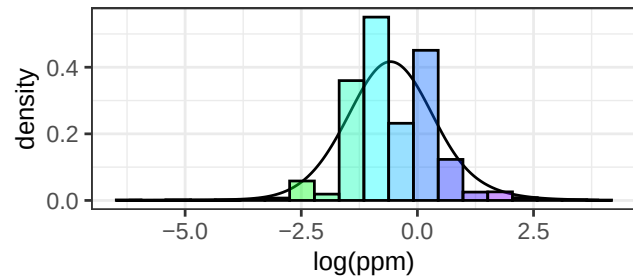
Data Entry Errors We identify data entry errors in the source variable, specifically two observations with values “fuck u” and “Eronhkkjhjjhjj”. These observations are removed from the dataset.

EDA

Response Variable (ppm)

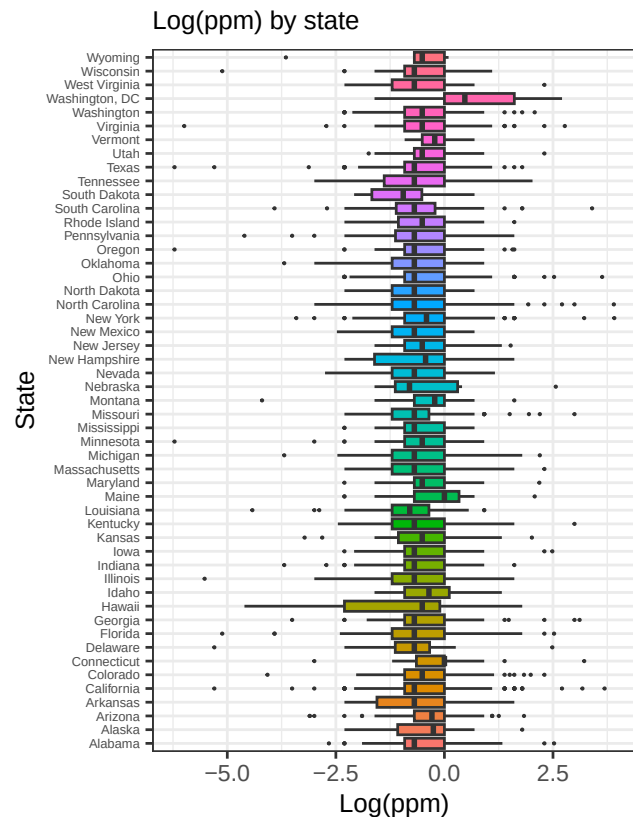
The distribution of ppm is highly right-skewed. We apply a log transformation to satisfy the normality assumption for the conditional distribution. The histogram of $\log(\text{ppm})$ below shows a relatively normal and symmetric distribution, making it suitable for our modeling approach.

Distribution of $\log(\text{ppm})$



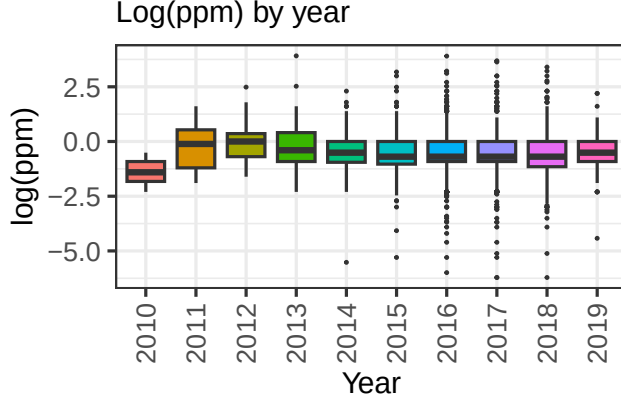
Random Effect

State & Region To account for heterogeneity, we assess `state` and `region` as potential random intercept. We see some variations by state, but not much in region. Therefore, we include a random intercept by state.



Fixed Effect

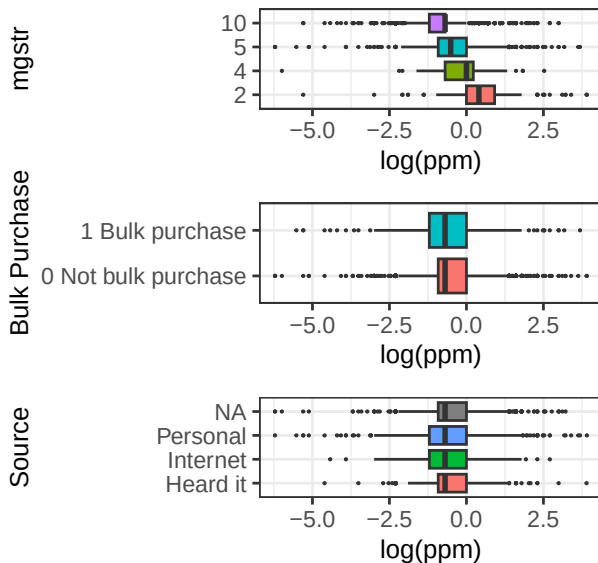
year The boxplot below shows variation in $\log(\text{ppm})$ across **year**, so including year as a fixed effect is also important to account for potential price inflation over time. We therefore include year in our model selection process.



mgstr, bulk_purchase, and source We next assess the relationship of **mgstr**, **bulk_purchase**, and **source** by $\log(\text{ppm})$ in the below boxplots. We see that all of these variables seem to have differences in $\log(\text{ppm})$ by their respective levels, thus we choose to include mgstr, bulk_purchase, and source as fixed effects in our model selection process.

Most mgstr (dosage strength) values are either 2, 4, 5, or 10 mg. We remove observations with other dosage values and code mgstr as a factor variable with four levels.

Log(ppm) by other Fixed Effect



Model Building

Model Selection We choose fixed effects for **year**, **mgstr**, **bulk_purchase** and **source**, also including a random intercept by state. For interaction terms, we run an exhaustive search on all models that include our desired fixed/main effects, random intercept for **state**, and all 2-way interactions. This gives $2^{10} = 1024$ models to select from, and we then select the model in the lowest BIC as our final model.

Table 1: Model Selection using BIC

model	BIC
$\log(\text{ppm}) \sim \text{mgstr} + \text{bulk_purchase} + \text{source} + (1 \mid \text{state})$	6922
$\log(\text{ppm}) \sim \text{year} + \text{mgstr} + \text{bulk_purchase} + \text{source} + (1 \mid \text{state})$	6929
$\log(\text{ppm}) \sim \text{mgstr} + \text{bulk_purchase} + \text{year}:\text{source} + (1 \mid \text{state})$	6929
$\log(\text{ppm}) \sim \text{year} + \text{mgstr} + \text{bulk_purchase} + \text{year}:\text{source} + (1 \mid \text{state})$	6929
$\log(\text{ppm}) \sim \text{mgstr} + \text{source} + \text{year}:\text{bulk_purchase} + (1 \mid \text{state})$	6929
$\log(\text{ppm}) \sim \text{year} + \text{mgstr} + \text{source} + \text{year}:\text{bulk_purchase} + (1 \mid \text{state})$	6929
$\log(\text{ppm}) \sim \text{mgstr} + \text{year}:\text{bulk_purchase} + \text{year}:\text{source} + (1 \mid \text{state})$	6929
$\log(\text{ppm}) \sim \text{year} + \text{mgstr} + \text{year}:\text{bulk_purchase} + \text{year}:\text{source} + (1 \mid \text{state})$	6929
$\log(\text{ppm}) \sim \text{year} + \text{bulk_purchase} + \text{source} + \text{year}:\text{mgstr} + (1 \mid \text{state})$	6929
$\log(\text{ppm}) \sim \text{bulk_purchase} + \text{source} + \text{year}:\text{mgstr} + (1 \mid \text{state})$	6929

Model with the lowest BIC is:

$$Y_{ij} = \mu + \alpha_j + \beta_1 \mathbb{I}(\text{mgstr}_{ij} = 4) + \beta_2 \mathbb{I}(\text{mgstr}_{ij} = 5) + \beta_3 \mathbb{I}(\text{mgstr}_{ij} = 10) + \beta_4 \mathbb{I}(\text{bulk}_{ij} = 1) + \beta_5 \mathbb{I}(\text{source}_{ij} = \text{internet}) + \beta_6 \mathbb{I}(\text{source}_{ij} = \text{personal}) + \epsilon_{ij},$$

$$\alpha_j \stackrel{iid}{\sim} \text{Normal}(0, \tau^2),$$

$$\epsilon_{ij} \stackrel{iid}{\sim} \text{Normal}(0, \sigma^2),$$

where Y_{ij} is the $\log(\text{ppm})$ for purchase i in state j , and mgstr_{ij} , bulk_{ij} , and source_{ij} are fixed effects. The reference level for mgstr is 2mg, for bulk purchase is 0 (non-bulk), and for source is "Heard of".

Estimation

We run both a frequentist and Bayesian hierarchical model to estimate the parameters.

Bayesian Model Fitting

Since we do not have much information about the model, we used non-informative priors.

$$\beta_j \stackrel{iid}{\sim} \text{Normal}(0, 1)$$

$$\tau^2 \sim \text{InvGamma}(0.1, 0.1)$$

$$\sigma^2 \sim \text{InvGamma}(0.1, 0.1)$$

Frequentist vs. Bayesian Model

In table 2, we compare the point estimates of the parameters from the frequentist model and posterior means of the parameters from the Bayesian model. We found that estimates from the Bayesian version of our model were very similar to the frequentist model we fitted, so we proceeded to interpret the frequentist model that we fitted above.

Table 2: Fixed effect estimates comparison

	Frequentist Estimate	Bayesian Estimate
(Intercept)	0.468	0.464
mgstr4	-0.553	-0.544
mgstr5	-0.793	-0.787
mgstr10	-1.254	-1.248
Bulk_purchase Bulk purchase	-0.199	-0.198
sourceInternet	-0.098	-0.098
sourcePersonal	-0.062	-0.062

Inference

Fixed effects

Table 3: Fixed effect estimates by Frequentist

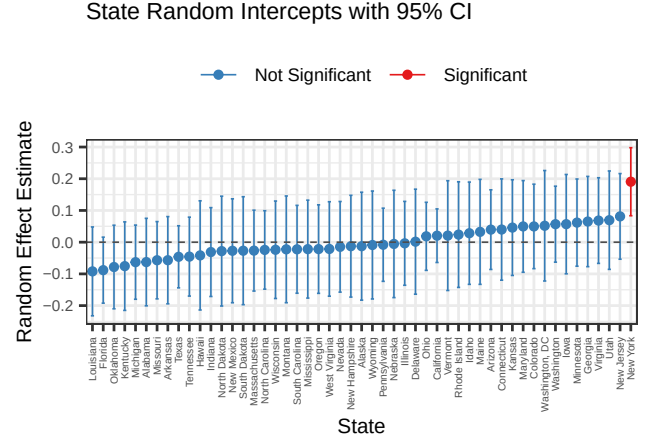
Std.Err	df	t	p
(Intercept)	0.468	0.058	8.135
mgstr4	-0.553	0.092	-6.026
mgstr5	-0.793	0.055	-14.421
mgstr10	-1.254	0.053	-23.824
bulk_purchase1 Bulk purchase	-0.199	0.035	-5.658
sourceInternet	-0.098	0.058	-1.686
sourcePersonal	-0.062	0.036	-1.710

All fixed effects are statistically significant at the 5% level except for **sourceInternet** and **sourcePersonal**. Since we modeled $\log(\text{ppm})$, we exponentiate coefficients to obtain price ratios:

- **Intercept:** The baseline ppm for the reference group (2mg dosage, non-bulk purchase, “Heard of” source) is $e^{0.468} = 1.60$ per mg.
- **Dosage strength:** Relative to 2mg pills, 4mg pills have a price ratio of $e^{-0.553} = 0.575$, 5mg pills have a ratio of $e^{-0.793} = 0.453$, and 10mg pills have a ratio of $e^{-1.254} = 0.285$. Higher dosages are significantly cheaper per mg.
- **Bulk purchase:** Bulk purchases have a price ratio of $e^{-0.199} = 0.820$ relative to non-bulk purchases.
- **Source:** Relative to “Heard of” reports, internet sources have a price ratio of $e^{-0.098} = 0.907$ and personal reports have a ratio of $e^{-0.062} = 0.940$, though these effects are not statistically significant.

Random effects

The following illustrates the sorted estimated random state intercepts with 95% confidence intervals.



The figure shows heterogeneity across different states. In particular, New York State does not contain 0 in its 95% confidence interval. It has a significant difference in its baseline ppm and the mean across all states.

Table 4: Intraclass Correlation Coefficient

Component	Value
State Variance (τ^2)	0.0082
Residual Variance (σ^2)	0.6658
ICC ($\tau^2/(\tau^2 + \sigma^2)$)	0.0121

The low ICC of 0.0121 indicates that only 1.2% of price variability is explained by state differences. The vast majority is attributable to within-state variance.

Model Diagnostics and Limitations

Variance Components

The with-in group variance is 0.6658. It suggests there is still a lot of within-group variance that the model could not explain.

Missing Data Considerations

We could also use MICE for missing data imputation.

Appendices

Code Appendix

```
knitr::opts_chunk$set(echo = FALSE, warning = FALSE, message = FALSE,
                      fig.align = 'center', fig.width = 3.5, fig.height = 2.5,
                      out.width = "\\columnwidth")
options(knitr.table.format = "latex")
library(tidyverse)
library(lme4)
library(rstan)
library(brms)
library(knitr)
library(kableExtra)
library(patchwork)
library(lubridate)
library(gridExtra)
library(influence.ME)

options(scipen = 0, digits = 4)
ggplot2::theme_set(ggplot2::theme_bw())
#load data
load("streetrx.RData")

# GROUP 5 - diazepam
# filter data for only diazepam
streetrx <- streetrx %>%
  filter(api_temp == "diazepam")
# variable description table
tibble(Variable = names(streetrx),
Description = c("Price per mg (outcome of interest)",
                "Year and quarter drug was purchased",
                "Date of the reported purchase",
                "city purchased",
                "state purchased",
                "country purchased",
                "northeast, midwest, west, south, or other/unknown",
                "source of information",
                "active ingredient of drug of interest, in our case Diazepam",
                "formulation of the drug (e.g., pill, patch, suppository)",
                "dosage strength in mg of the units purchased",
                "indicator for purchase of 10+ units at once",
                "primary reason for purchase"
)) %>%
  kable(caption = "Variable Descriptions", booktabs = TRUE)
# code missing data
streetrx <- data.frame(apply(streetrx, 2, function(x) gsub("^$|^$", NA, x)))
```

```

# table of variables with missing data
# tibble(
#   Variable = names(streetrx),
#   `Number Missing` = apply(streetrx, 2, function(x) sum(is.na(x))),
#   `Proportion Missing` = apply(streetrx, 2, function(x) mean(is.na(x)))
# ) %>%
#   kable(caption = "Number of observations missing per variable") %>%
#   kable_styling(latex_options = "HOLD_position")

# code factors and numeric variables
streetrx <- streetrx %>%
  mutate(ppm = as.numeric(ppm),
         yq_pdate = as.numeric(yq_pdate),
         price_date = mdy(price_date),
         city = as.factor(city),
         state = as.factor(state),
         country = as.factor(country),
         USA_region = as.factor(USA_region),
         source = as.factor(source),
         form_temp = as.factor(form_temp),
         mgstr = as.numeric(mgstr),
         bulk_purchase = as.factor(bulk_purchase),
         Primary_Reason = as.factor(Primary_Reason)
  )

# add year
streetrx <- streetrx %>%
  mutate(year = year(price_date))

# delete observations with missing ppm data
streetrx <- streetrx %>%
  filter(!is.na(ppm))

# remove data entry errors in source variable
streetrx <- streetrx %>%
  filter(!(source %in% c("fuck u", "Eronhkkjhjhjhj")))

# delete levels of mgstr that are not 2, 4, 5, 10
streetrx <- streetrx %>%
  filter(mgstr %in% c(2,4,5,10)) %>%
  mutate(mgstr = as.factor(mgstr))

# delete year of 1969 as it is likely an error
streetrx <- streetrx %>%
  filter(year != 1969)

# delete state = USA
streetrx <- streetrx %>%
  filter(state != "USA")

#combine all website sources as being Internet source
streetrx <- streetrx %>%
  mutate(source = as.character(source)) %>%
  mutate(source = if_else(str_detect(source, "http://"), "Internet", source)) %>%
  mutate(source = if_else(str_detect(source, ".com$"), "Internet", source)) %>%
  mutate(source = if_else(source == "Streetrx", "Internet", source)) %>%

```

```

mutate(source = if_else(source == "Poopy,", "N/A", source)) %>%
mutate(source = if_else(source == "google", "Internet", source)) %>%
mutate(source = if_else(source == "Internet Pharmacy", "Internet", source)) %>%
mutate(source = na_if(source, "N/A")) %>%
mutate(source = na_if(source, "None")) %>%
mutate(source = as.factor(source))

# remove states with 1 observation
streetrx <- streetrx %>%
  group_by(state) %>%
  count() %>%
  filter(n > 1) %>%
  select(state) %>%
  left_join(streetrx, by = "state")
streetrx %>%
  ggplot(aes(x = log(ppm), y = ..density..)) +
  geom_density(color = "black", adjust = 5) +
  geom_histogram(alpha = 0.4, fill=rainbow(20), bins=20, color = "black") +
  labs(title = "Distribution of log(ppm)") +
  theme(plot.title = element_text(size = 10),
        axis.title = element_text(size = 10))
ggplot(streetrx, aes(x = log(ppm), y = state)) +
  geom_boxplot(aes(fill = factor(state)), outlier.size = 0.1) +
  labs(title = "Log(ppm) by state",
        y = "State",
        x = "Log(ppm)") +
  theme(legend.position = "none",
        axis.text.y = element_text(size = 5),
        plot.title = element_text(size = 10),
        axis.title = element_text(size = 10))
streetrx %>%
  mutate(
    year = yq_pdate %/% 10,
    quarter = yq_pdate %% 10
  ) %>%
  filter(year > 2009) %>%
  ggplot(., aes(x = factor(year), y = log(ppm))) +
  geom_boxplot(aes(fill = factor(year)), outlier.size = 0.1) +
  labs(title = "Log(ppm) by year",
        x = "Year") +
  theme(legend.position = "none",
        axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1),
        plot.title = element_text(size = 10),
        axis.title = element_text(size = 10))
mgstr_ppm <- ggplot(streetrx, aes(x = factor(mgstr), y = log(ppm))) +
  geom_boxplot(aes(fill = factor(mgstr)), outlier.size = 0.1) +
  labs(x = "mgstr") +
  theme(legend.position = "none",
        axis.title = element_text(size = 10)) +
  coord_flip()

bp_ppm <- ggplot(streetrx, aes(x = bulk_purchase, y = log(ppm))) +
  geom_boxplot(aes(fill = bulk_purchase), outlier.size = 0.1) +
  labs(x = "Bulk Purchase") +
  theme(legend.position = "none",
        axis.title = element_text(size = 10)) +

```

```

coord_flip()

source_ppm <- ggplot(streetrx, aes(x = source, y = log(ppm))) +
  geom_boxplot(aes(fill = source), outlier.size = 0.1) +
  labs(x = "Source") +
  theme(legend.position = "none",
        axis.title = element_text(size = 10)) +
  coord_flip()

patchwork <- mgstr_ppm / bp_ppm / source_ppm

patchwork + plot_annotation(
  title = "Log(ppm) by other Fixed Effect",
  theme = theme(plot.title = element_text(size = 10))
)

exhaustive_search <- function(raw_model, vars, data, REML = F) {
  y_name <- deparse(raw_model[[2]])
  group_name <- deparse(raw_model[[3]])
  id <- 0 : (2^length(vars) - 1)
  construct_model <- function(.id){
    subset <- (.id %/% 2^(0:(length(vars) - 1))) %% 2 == 1
    if (all(subset == F)){
      RHS <- paste0(c("1", group_name), collapse = ' + ')
    }
    else {
      RHS <- paste0(c(vars[subset], group_name), collapse = ' + ')
    }
    paste(y_name, RHS, sep = ' ~ ')
  }
  run_model <- function(.id){
    model_str <- construct_model(.id)
    model_formula <- as.formula(model_str)
    res <- lmer(model_formula, REML = REML, data = data)
    return (summary(res)$AICtab)
  }

  bind_cols(
    model = sapply(id, construct_model),
    as.data.frame(t(sapply(id, run_model)))
  )
}

ex_result_int <- exhaustive_search(
  raw_model <- log(ppm) ~ (1 | state),
  vars = c("year", "mgstr", "bulk_purchase", "source",
           "year:mgstr", "year:bulk_purchase", "year:source",
           "mgstr:bulk_purchase", "mgstr:source", "bulk_purchase:source"),
  data = streetrx,
  REML = F
)

ex_result_int %>% arrange(BIC) %>% head(10) %>%
  select(1, 3) %>%
  kbl(caption = "Model Selection using BIC", booktabs = TRUE) %>%
  kable_styling(font_size = 10, latex_options = c("HOLD_position", "scale_down"))

best_model <- ex_result_int %>% arrange(BIC) %>% .[1, "model"]
res <- lmer(as.formula(best_model), REML = F, data = streetrx)
summary(res)

```

```

priors <- c(
  set_prior("normal(0, 1)", class = 'b'),
  set_prior("inv_gamma(0.1, 0.1)", class = "sd", group = "state"),
  set_prior("inv_gamma(0.1, 0.1)", class = 'sd')
)

bayes_result <- brm(as.formula(best_model), data = streetrx, prior = priors,
  verbose = F, refresh = 0)

bayes_fixed <- (summary(bayes_result)$fixed) %>%
  select("Estimate") %>%
  rename("Bayesian Estimate" = "Estimate")

freq_est <- as.matrix(coef(summary(res))[,1])
colnames(freq_est)[[1]] <- "Frequentist Estimate"

# Simplify row names for better display
rownames(freq_est) <- gsub("bulk_purchase1", "Bulk_purchase", rownames(freq_est))

cbind(freq_est, bayes_fixed) %>%
  knitr::kable(caption = "Fixed effect estimates comparison", longtable = FALSE, digits = 3) %>%
  kable_styling(font_size = 10, latex_options = c("HOLD_position", "scale_down"))

bayesian_var <- rbind(summary(bayes_result)$spec_pars %>%
  select("Estimate"),
  summary(bayes_result)$random$state %>% select("Estimate")) %>%
  mutate(Estimate = Estimate^2) %>%
  rename("Bayesian Model Estimate" = "Estimate")

rownames(bayesian_var) <- c("Residual", "State")

frequentist_est <- as.data.frame(VarCorr(res)) %>%
  select(1, 4) %>%
  rename(var = vcov) %>%
  rename("Frequentist Model Estimate" = var) %>%
  select("Frequentist Model Estimate")

rownames(frequentist_est) <- c("State", "Residual")

# merge(bayesian_var, frequentist_est, by = "row.names") %>%
#   rename("Source of Variation" = "Row.names") %>%
#   knitr::kable(caption = "Variance estimates comparison", longtable = FALSE, digits = 4) %>%
#   kable_styling(font_size = 7, latex_options = c("HOLD_position", "scale_down"))
coef(summary(res)) %>%
  knitr::kable(caption = "Fixed effect estimates by Frequentist", longtable = FALSE, digits = 3,
    col.names = c("Est.", "Std.Err", "df", "t", "p")) %>%
  kable_styling(font_size = 9, latex_options = c("HOLD_position", "scale_down"))
# dotplot - horizontal with state on x-axis
ranef_data <- ranef(res, condVar = TRUE)$state
se_data <- sqrt(attr(ranef_data, "postVar")[1, 1, ])
state_effects <- data.frame(
  state = rownames(ranef_data),
  estimate = ranef_data[, 1],
  se = se_data
) %>%

```



```

mutate(
  lower = estimate - 1.96 * se,
  upper = estimate + 1.96 * se,
  significant = ifelse(lower > 0 | upper < 0, "Significant", "Not Significant")
) %>%
arrange(estimate)

ggplot(state_effects, aes(x = reorder(state, estimate), y = estimate, color = significant)) +
  geom_point(size = 1.2) +
  geom_errorbar(aes(ymin = lower, ymax = upper), width = 0.3, linewidth = 0.3) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "gray30", linewidth = 0.3) +
  scale_color_manual(values = c("Significant" = "#E41A1C", "Not Significant" = "#377EB8")) +
  labs(x = "State", y = "Random Effect Estimate",
       title = "State Random Intercepts with 95% CI") +
  theme_bw() +
  theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust = 1, size = 4),
        axis.text.y = element_text(size = 6),
        axis.title = element_text(size = 8),
        plot.title = element_text(size = 9),
        legend.position = "top",
        legend.title = element_blank(),
        legend.text = element_text(size = 7))
# Calculate ICC
var_components <- as.data.frame(VarCorr(res))
tau_sq <- var_components$vcov[1] # state variance
sigma_sq <- var_components$vcov[2] # residual variance
icc <- tau_sq / (tau_sq + sigma_sq)

data.frame(
  Component = c("State Variance ( $\tau^2$ )", "Residual Variance ( $\sigma^2$ )", "ICC ( $\tau^2/(\tau^2 + \sigma^2)$ )"),
  Value = c(tau_sq, sigma_sq, icc)
) %>%
knitr::kable(caption = "Intraclass Correlation Coefficient", longtable = FALSE, digits = 4, escape = FALSE)
kable_styling(font_size = 9, latex_options = c("HOLD_position"))

```